

The empirical recurrence for OEIS A252721, proved by transfer matrix

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Abstract

OEIS A252721 counts arrays of width 4 over an alphabet of 4 letters in which every three consecutive entries taken along a row, a column or a diagonal hold a prescribed number of DISTINCT values, the arrays being counted up to renaming of the letters. The entry carries an empirical recurrence of order 5 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Nothing in the condition compares letters by size, so the count up to renaming is recovered from counts over fixed alphabets by inverting a triangular system of falling factorials; each of those is a walk count on pairs of consecutive rows, and the same symmetry is used a second time to run the walk on ORBITS of pairs, which is what brings the state count down to $S = 131$.

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1 The conjecture

OEIS A252721 is “Number of $(2+2) \times (n+2)$ 0..3 arrays with every consecutive three elements in every row and diagonal having exactly two distinct values, and in every column and antidiagonal not having exactly two distinct values, and new values 0 upwards introduced in row major order.”. It has offset 1 and begins

45, 46, 50, 63, 69, 91, 107, 131, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = a(n-2) + 2*a(n-3) - 2*a(n-5)$ for $n > 8$.

As of the “Last modified” line on the live entry (Dec 05 2018 15:41:01, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has width 4 and entries in $\{0, 1, \dots, 3\}$. Four families of triples run through it, and each carries its own condition on how many distinct values the three entries may hold:

along	the triple	admissible numbers of distinct values
a row	$(i,j), (i,j+1), (i,j+2)$	1, 3
a column	$(i,j), (i+1,j), (i+2,j)$	2
a diagonal (nw-se)	$(i,j), (i+1,j+1), (i+2,j+2)$	2
an antidiagonal (ne-sw)	$(i,j), (i+1,j-1), (i+2,j-2)$	1, 3

A triple counts only when all three of its cells lie inside the array. The entry writes the array with n as the number of COLUMNS. The walk is run down the transpose, which exchanges rows

with columns and carries each diagonal direction to itself, so the two conditions are exchanged once and the diagonal ones left alone.

The clause “new values 0 upwards introduced in row major order” says the array is the canonical representative of its EQUALITY PATTERN: reading the cells in row major order, a letter may appear for the first time only if every smaller letter already has. So what the entry counts is patterns, not arrays.

3 From patterns to fixed alphabets

Write N_j for the number of admissible patterns using exactly j letters and L_i for the number of admissible arrays over an alphabet of i letters. An array over i letters is a pattern together with an injection of its letters into the alphabet, so

$$L_i = \sum_j N_j i(i-1)\cdots(i-j+1),$$

a triangular system with unit diagonal, hence invertible over $\mathbf{Q}$. The entry counts $a = N_1 + \cdots + N_4$, so

$$a = \sum_{i=1}^4 w_i L_i, \quad w = \mathbf{1}^\top F^{-1},$$

F being the matrix of falling factorials.

This step is legitimate only if the condition is invariant under permuting the alphabet, and here it is: every clause counts how many DISTINCT values a triple holds, and whether two cells agree is untouched by any renaming. No clause compares letters by size and none does arithmetic on them, so the reduction applies with nothing left over.

4 Each L_i is a walk count, on orbits

Every triple lies inside three consecutive rows, so with $c_1, \dots, c_R$ the rows of the array the condition is a conjunction of tests on the windows (c_t, c_{t+1}, c_{t+2}) together with the row test on each c_t separately. A pair of consecutive admissible rows is therefore a state, and L_i counts walks in the digraph on those pairs, one edge $(c_t, c_{t+1}) \rightarrow (c_{t+1}, c_{t+2})$ for each admissible window.

That digraph has up to $i^{2 \cdot 4}$ vertices, which is more than can be written down for the wider arrays. The symmetry is used again to cut it. Let $G = S_i$ act on states by renaming letters. Every test is invariant, so the adjacency matrix M commutes with the action, and the all-ones vector τ is invariant; hence $M^t \tau$ is invariant, that is, constant on orbits. Writing $\bar{M}[O, O']$ for the number of edges from one fixed representative of O to states of O' — a number independent of the representative, again by invariance — gives $\overline{M^t \tau} = \bar{M}^t \bar{\tau}$, and therefore

$$L_i(R) = \sum_s (M^{R-2} \tau)(s) = \sum_O |O| (\bar{M}^{R-2} \bar{\tau})(O),$$

the sum being over the orbits of admissible pairs. An orbit is exactly the PATTERN of the pair, and if that pattern uses p letters its orbit has $i(i-1)\cdots(i-p+1)$ members, since a state is fixed only by permutations fixing every letter it shows. So the start vector is the falling factorial and the end vector is all ones.

Assembling the $i = 1, \dots, 4$ pieces as a disjoint union, with ι carrying the weights w_i cleared of denominators,

$$a(n) = \iota^\top M^j \tau,$$

with $S = 131$ states in all. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

5 The proof

Lemma 1. *Let $q(t) = t^5 - \sum_i c_i t^{5-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+5} - \sum_i c_i M^{j+5-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = a(n-2) + 2a(n-3) - 2a(n-5)$$

for every $n > 8$.

Proof. The vector $w = q(M)\tau$ was formed by 5 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 8 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The rational weights w_i are carried as one common denominator throughout, so no division is performed until the final term. $\square$

6 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 43 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: fill the cells in row major order, allow a cell to introduce a new letter only when every smaller letter is already present, and reject a partial array as soon as a triple all of whose cells are placed violates its condition. This uses neither the transfer matrix, nor the orbit reduction, nor the falling-factorial inversion, and it returns the published terms.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*, Springer, 1976. (Lumping a chain along a partition preserved by the transition rule, Section 6.3.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.